

Practical 8 - Matched studies and their analysis

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `vit_E.csv`

Introduction

The aim of this practical is to consolidate the ideas of matching discussed in the lecture. We will use data from a matched case-control study of the association between vitamin E measured in the blood and occurrence of cancer. The dataset is called `vit_E.csv`.

High levels of vitamin E are thought by some to be protective against cancer. This hypothesis was investigated by measuring Vitamin E in stored blood samples from 271 men in a large cohort study initially aged 35–64 years who subsequently developed cancer. These values were compared with those from control men who had not, at that time, developed cancer. One control for each case was selected randomly from within the cohort study, subject to matching for age (within 5 years), duration of storage of the blood sample (within 3 months) and smoking status.

Note that this dataset does not include the matching variables. It does include ‘observation time’, which is the time from blood collection to diagnosis of cancer in the cases. In this session we will analyse the data assuming no confounding.

Variable	Description
<code>set</code>	Case-control unique identifier taking values from 1 to 271
<code>case</code>	0: control, 1: case
<code>vitE</code>	Vitamin E (mg/dl)

Variable	Description
time	Observation time: 1: up to 1 year, 2: 1–3 years, 3: over 3 years

Aims

- Understand how to appropriately conduct hypothesis tests on data from a matched case-control study.
- Use appropriate statistical tests to analyse matched studies with continuous dependent variables.

Analysis

Question 1

Load the data and take a look at the first few rows to ensure you understand how it is structured.

SOLUTION

```
vit <- read.csv("vit_E.csv")
head(vit)
```

```

      set case time vitE
1      1     1    1 10.2
2      1     0    1 18.2
3      2     1    3 15.3
4      2     0    3 11.1
5      3     1    2  9.7
6      3     0    2 10.8
```

We can see that there two rows from each set (or block). The first row is the data for the case, and the second for the control. The data are already sorted by **set** (and then **case**.)

Question 2

Summarize the vitamin E measurements for cases and controls. What is the difference between the means in the two groups? Use suitable plots to display vitamin E levels in the cases and controls.

SOLUTION

```
summary(vit$vitE[vit$case==1])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	7.80	9.70	10.03	11.85	33.50

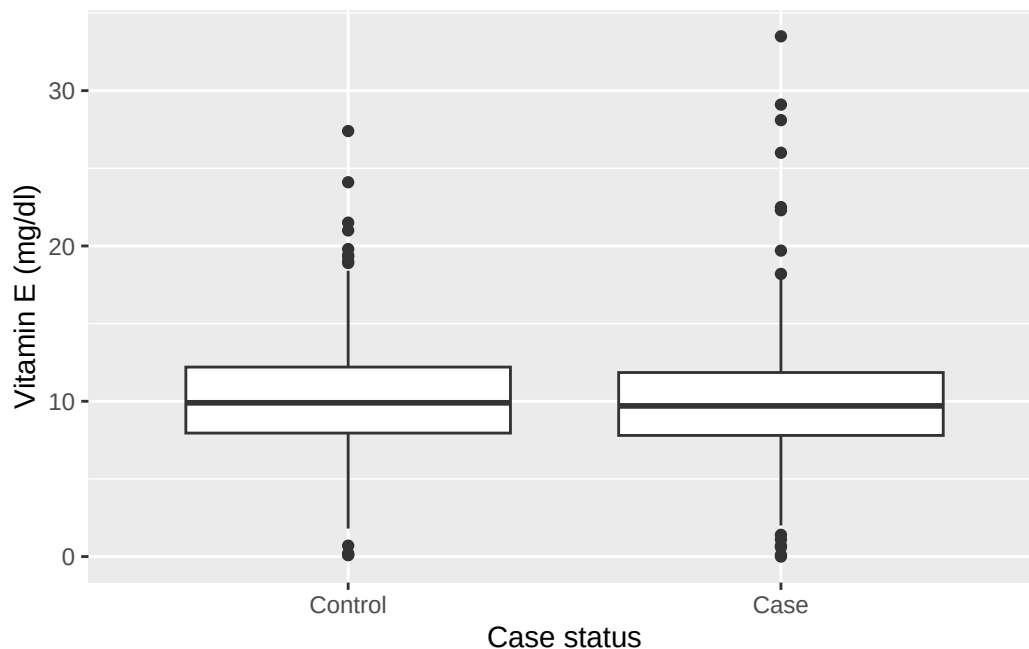
```
summary(vit$vitE[vit$case==0])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.10	7.95	9.90	10.18	12.20	27.40

The mean vitamin E level is a little higher in controls compared with cases.

To plot we first try a box plot:

```
vit$case <- factor(vit$case,  
                  labels=c("Control","Case"))  
library(ggplot2)  
ggplot(vit, aes(x = case, y = vitE)) +  
  geom_boxplot() +  
  labs(  
    x = "Case status",  
    y = "Vitamin E (mg/dl)"  
  )
```



A drawback of this plot is that it does not capture the matched nature of the data. That is, we cannot visually see the pair-wise differences. An alternative that would overcome this would be to plot the distribution of case-control differences.

Question 3

For questions 3-5 we will analyse these data treating vitamin E as the dependent variable, and cancer as the independent variable. In reality the vitamin E reading is best thought of as an exposure since blood was taken and stored before the onset of cancer.

We will first analyse these data using linear regression.

We first demonstrate an invalid approach, to demonstrate the importance of allowing for block in any analysis. Fit a linear regression model with just the case-control variable as an explanatory variable.

```
lm1 <- lm( vitE ~ case, data=vit)
summary(lm1)
```

Now add the set variable, as a factor covariate in the regression model.

```
lm2 <- lm( vitE ~ case + factor(set), data=vit)
summary(lm2)
```

Compare the estimates of the case-control coefficient and its standard error from the two analyses. Is there evidence that people who developed cancer had different vitamin E levels than people who did not develop cancer?

SOLUTION

```
lm1 <- lm( vitE ~ case, data=vit)
summary(lm1)
```

Call:

```
lm(formula = vitE ~ case, data = vit)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.0771	-2.2306	-0.2771	1.8694	23.4694

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.1771	0.2403	42.356	<2e-16 ***
caseCase	-0.1465	0.3398	-0.431	0.667

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.955 on 540 degrees of freedom

Multiple R-squared: 0.0003441, Adjusted R-squared: -0.001507

F-statistic: 0.1859 on 1 and 540 DF, p-value: 0.6666

```
lm2 <- lm( vitE ~ case + factor(set), data=vit)
#there are lots of set coefficients. Here we just show
#the output relating to the intercept and case effect
summary(lm2)$coefficients[1:2,]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.2732477	2.5301318	5.6413061	4.256055e-08
caseCase	-0.1464945	0.3068235	-0.4774551	6.334244e-01

We can see that the estimated mean difference in Vitamin E levels between cases and controls is the same in each model, but the standard error has (appropriately) decreased once we account for the paired nature of the data.

The estimate shows that in our dataset we observed slightly lower vitamin E levels among people who later went on to be diagnosed with cancer. However, the p-value (from the second model) is nowhere near significant ($p = 0.63$).

Question 4

In order to make the data suitable for analysis with a paired t-test (and other analyses), we need to reconfigure it so that the data from each pair is contained in one row. This can be done using the reshape function, which is an extremely useful command, but can be fiddly to get to do what you want it to do.

Use the following line to reshape the data to wide form:

```
vitWide <- reshape(vit, v.names=c("vitE"), timevar="case",
                    idvar="set",direction="wide")
```

Then use `head` to check how `vitWide` looks.

SOLUTION

```
vitWide <- reshape(vit, v.names=c("vitE"), timevar="case",
                    idvar="set",direction="wide")
head(vitWide)
```

	set	time	vitE.Case	vitE.Control
1	1	1	10.2	18.2
3	2	3	15.3	11.1
5	3	2	9.7	10.8
7	4	3	10.9	10.7
9	5	3	5.3	12.1
11	6	2	12.5	12.9

The data from each set are now contained in one row. `reshape` has created the variables `vitE.Case` and `vitE.Control` which contain the vitamin E level of the case and control respectively.

Question 5

Using the data in wide format, carry out a paired t-test of the null hypothesis that the mean blood vitamin E levels are the same in cases and controls.

How do the estimated mean difference and its standard error compare to those from the linear regression models above?

Carry out some assessment of the assumptions made by the paired t-test.

SOLUTION

```
t.test(vitWide$vitE.Control, vitWide$vitE.Case, paired = T)
```

Paired t-test

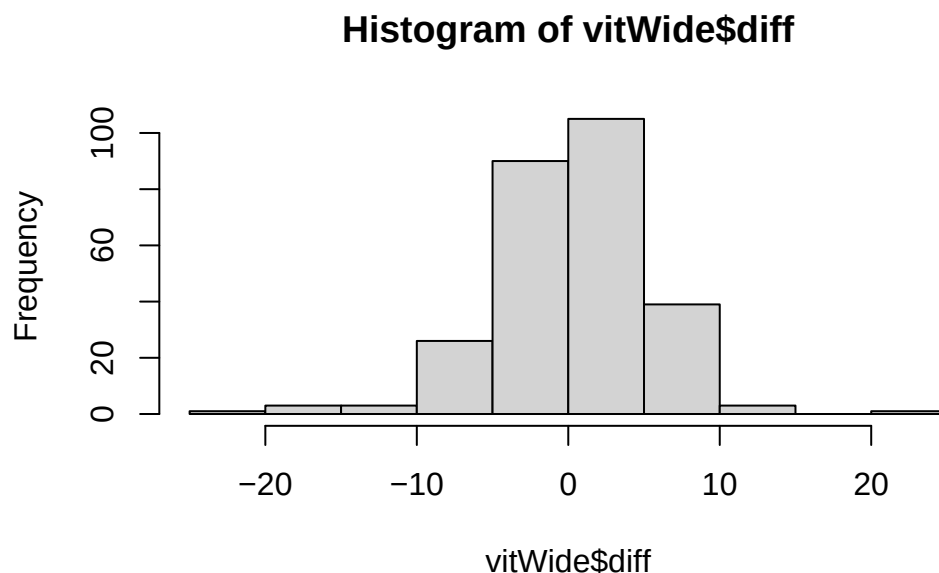
```
data: vitWide$vitE.Control and vitWide$vitE.Case
t = 0.47746, df = 270, p-value = 0.6334
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.4575764  0.7505653
sample estimates:
mean difference
 0.1464945
```

The null hypothesis here is that the mean difference in vitamin E levels between cases and controls is zero, which is equivalent to stating that the mean vitamin E levels are the same in cases and controls. Here the p-value is 0.63 - we have no evidence against this null hypothesis.

The estimated mean difference and its standard error here are identical to those from the regression model that included matched set as a covariate (`lm2`).

The paired t-test is valid provided the paired differences are normally distributed, or if the sample size is sufficiently large for the central limit theorem to apply. We can examine the distribution of the paired differences with:

```
vitWide$diff <- vitWide$vitE.Control-vitWide$vitE.Case
hist(vitWide$diff)
```



Here, the distribution appears fairly normal, though we do see a few outliers that are more extreme than would be expected for a normal distribution. Having said that, the sample size is 271 (pairs), which is quite large, so we would probably be quite happy to use the (paired) t-test here.

Question 6

For the remainder of this session we will dichotomise vitamin E into a binary variable and perform analysis to estimate the odds ratio of association. Here we can use the reversible nature of an odds ratios so that we can interpret vitamin E as the exposure and cancer as the outcome or vise-versa.

In `vitWide`, create variables `vitECaseHigh` and `vitEControlHigh` that indicate whether the case (respectively control) have a vitamin E level above 12mg/dl. Then cross-tabulate these two variables.

SOLUTION

```
vitWide$vitECaseHigh <- 1*(vitWide$vitE.Case>12)
vitWide$vitEControlHigh <- 1*(vitWide$vitE.Control>12)
table(vitWide$vitECaseHigh,vitWide$vitEControlHigh)
```

	0	1
0	159	54
1	41	17

Question 7

Using the table generated in the previous question, perform McNemar's test using `binom.test`.

SOLUTION Using the notation from the lecture notes we have $q = 159, r = 54, s = 41, t = 17$. The null hypothesis is that the expectation of the numbers of discordant pairs of the two types (r, s) are the same, or in other words that S arises from a binomial distribution of size $r + s$ with expectation $(r + s)/2$. If the null hypothesis is true then the number of discordant pairs with the case exposed (high vitamin E) and the control unexposed, observed here to be $s = 41$, is distributed $Binomial(r + s, 0.5)$. We may refer to s (or r for that matter) as the test statistic.

We can thus perform McNemar's test using

```
binom.test(41, 95)
```

Exact binomial test

```
data: 41 and 95
number of successes = 41, number of trials = 95, p-value = 0.2181
alternative hypothesis: true probability of success is not equal to 0.5
95 percent confidence interval:
 0.3303129 0.5372285
sample estimates:
probability of success
      0.4315789
```

Note that there is an argument p for `binom.test` for specifying the null probability we wish to test. By not specifying a value for it, it has defaulted to 0.5 (which is what we want here).

The p-value for the 2-sided test is 0.22 - there is no evidence against the null hypothesis.

Question 8

Use the table you generated in question 6 and the output from `binom.test` in question 7 to estimate the odds ratio and a 95% CI for it.

SOLUTION

The odds ratio estimate is given by

41/54

```
[1] 0.7592593
```

We can construct a 95% CI for the OR by transforming the CI limits obtained earlier from `binom.test`:

```
0.3303129/(1-0.3303129)
```

```
[1] 0.4932347
```

```
0.5372285/(1-0.5372285)
```

```
[1] 1.160894
```

Question 9

What are your epidemiological conclusions from the analysis where vitamin E status is dichotomised? Working together with one or more colleagues, write a short paragraph to summarise your findings.

SOLUTION

The interpretation of the OR is that the odds of high vitamin E status in cases is estimated to be 0.76 times the odds of high vitamin E status in controls conditional on age, duration of sample storage and smoking status (the matching variables). This is equivalent to saying that the odds of being a case among those with high vitamin E status is estimated to be 0.76 times the odds of being a case among those with low vitamin E status conditional on age, duration of sample storage and smoking status. Alternatively, one could say that the odds of being a case is estimated to be 24% lower in those with high vitamin E status than in those with low vitamin E status conditional on age, duration of sample storage and smoking status. The confidence interval for the OR contains the value 1: therefore, there is no evidence (at the 5% level) against the null hypothesis that there is no association between case-control status and vitamin E status.

Summary: Adjusted for age, duration of sample storage and smoking status there was no evidence of an association between high vitamin E status (> 12 mg/dl) and subsequent risk of cancer ($p=0.22$). Among those with the same age, duration of sample storage and smoking status, those with high vitamin E status had odds of cancer that were 24% lower (95% CI 51% lower to 16% higher) than those with low vitamin E status (odds ratio = 0.76 (95% CI 0.49 to 1.16)).